

Thesis Goal:

Automatically extend a seed training set for entity matching with useful additional training pairs. Evaluate the utility of the training sets by using it to fine-tune DITTO or Roberta and apply the fine tuned model to the test set afterwards.

Result: Knowledge about which data augmentation technique or combination of techniques works best

Data Augmentation Techniques to try:

1. String-based Data Augmentation (DITTO style)
2. Use LLM to label additional pairs, e.g. use training set, generate more pairs by recombining entity descriptions from the training set using some blocking technique, have LLM label interesting pairs, e.g. maybe using active learning (Aaron's Paper, attached)
3. Use Web RAG (Tavily) to discover additional offers, e.g. search for offers using an entity description or its title as search term. Afterwards, form pairs combining seed description and retrieved offer, have LLM label new pairs as matches or non-matches.

Evaluation Datasets to use (please use at least 2 of them):

- WDC Products: Small Training Set, 80% CC
- Abt-Buy, Walmart-Amazon (as alternative product datasets)
- DBLP Scholar (as a dataset from a different domain)

References:

WDC Products - <https://webdatacommons.org/largescaleproductcorpus/wdc-products/>

Active Learning Code for LLM Pair labeling -

<https://github.com/wbsg-uni-mannheim/automatic-data-integration>

DITTO Code - <https://github.com/megagonlabs/ditto>

Link to Evaluation Datasets -

<https://github.com/anhaidgroup/deepmatcher/blob/master/Datasets.md>

Lecture slide set on Entity Matching -

https://www.uni-mannheim.de/media/Einrichtungen/dws/Files_Teaching/Web_Data_Integration/HWS2025/WDI05-IdentityResolution-HWS2025.pdf